

Does Learning a Wavelet Help?

Perfect-Reconstruction Constraints Enable Learnable Wavelets for Low-Dose CT Denoising

Author One*, Author Two*

*Affiliation

City, Country

Email

Abstract—Making the wavelet transform learnable is a natural extension of wavelet-domain image denoising. We show that for low-dose CT, this extension fails by default: an unconstrained learnable analysis-synthesis pair drifts away from invertibility during training, and the resulting network performs no better than a fixed Haar basis (-0.03 dB, $p = 0.11$, 95% CI straddling zero). We trace this to the transform’s reconstruction error, which reaches 12–15% after training, and show it is causal rather than merely correlated: injecting a controlled error of the same magnitude into the perfect-reconstruction arm degrades it to the fixed-Haar level. Imposing perfect reconstruction by construction — parameterizing the wavelet as a lifting scheme whose inverse shares the forward filters — makes learning pay off: $+0.30$ dB over unconstrained learning ($p = 0.0005$) and $+0.27$ dB over fixed Haar ($p = 0.0004$), consistent across two data splits. The resulting network has 2,159 parameters and reaches 31.97 dB on AAPM-Mayo 2016, within 1.0 dB of a RED-CNN baseline trained under the identical protocol at $857\times$ fewer parameters.

Index Terms—low-dose CT, image denoising, learnable wavelet, lifting scheme, perfect reconstruction, invertibility

I. Introduction

Low-dose CT (LDCT) reduces radiation exposure at the cost of increased noise. Deep denoising networks such as RED-CNN [1] are effective but parameter-heavy ($\sim 10^6$ parameters), which limits deployment on the resource-constrained workstations common in smaller clinical sites. Compact models are therefore attractive.

Wavelet-domain processing is a classical way to separate noise from structure, and recent work has made the wavelet itself learnable to adapt to a specific modality and dose [17], [18], [19], [20], [21], [22]. This paper asks a simple question:

Does making the wavelet learnable actually help?

Our answer, established through a controlled three-arm study, is: not by itself. An unconstrained learnable analysis-synthesis pair performs statistically indistinguishably from a fixed Haar basis (Section V-B). The reason is not that adaptation is useless, but that learning destroys the transform’s own invertibility: after training, the analysis-synthesis round-trip error reaches 12–15% of the signal norm, so the network must spend capacity compensating for a loss it should not have to model.

We then show that this is fixable by construction. Parameterizing the wavelet as a lifting scheme [23], [24] — in which the inverse is not a second set of parameters but the forward prediction/update filters applied in reverse — guarantees invertibility regardless of what the filters learn. With this constraint, learning finally pays off: $+0.30$ dB over unconstrained learning and $+0.27$ dB over fixed Haar, both highly significant.

Contributions.

- 1) A negative result: an unconstrained learnable wavelet gives no benefit over a fixed Haar basis for LDCT denoising.
- 2) A mechanism, established by intervention: the failure is caused by loss of invertibility. Injecting a controlled round-trip error of the same magnitude into the PR arm degrades it to the fixed-Haar level — monotonically and significantly ($n = 5$ seeds, $p \leq 0.007$ at every dose). Prior invertible-wavelet denoisers cannot observe this state, because their designs make it unreachable.
- 3) A practical caveat: the structural PR guarantee does not enforce itself in floating point. Bounding the lifting taps is what makes it hold — without it the round-trip error degrades from 3.6×10^{-7} to 5.2×10^0 , which the invertible-network literature does not report.
- 4) A 2,159-parameter network that approaches published baselines.

II. Related Work

A. Low-dose CT denoising

Convolutional encoder-decoders such as RED-CNN [1] established the field, followed by adversarial [2], [3] and, more recently, transformer- and diffusion-based methods [4], [5], [6], [7], [8], [9], [10], [11]. These improve fidelity but at 10^5 – 10^7 parameters. A parallel line targets efficiency instead: MLAR-UNet [13], CT-Mamba [15], AMFA-Net [16], FMDNet [14] and MobileMamba-UNet [12] all report competitive results at far lower cost, the last of these also operating in a wavelet domain. We share the efficiency goal but ask a different question — not how

small can a denoiser be, but does making the transform itself learnable help at all.

B. Wavelet-domain and multiresolution denoising

Wavelet shrinkage [27], [28] separates noise from structure by thresholding subband coefficients. Multi-level wavelet CNNs [17], [18] fold this structure back into deep networks, and recent LDCT methods pair wavelets with diffusion [11] or state-space models [12]. In all of these the transform is a fixed operator; the learnable capacity sits in the network around it.

C. Learnable transforms and invertibility

A separate line makes the transform itself learnable. LINN [19] learns lifting prediction/update steps around a fixed undecimated Haar transform; WINNet [20] additionally learns the splitting operator, constrained to be orthogonal via a Cayley transform so that PR is preserved by construction. Both therefore build invertibility into the architecture. Classical filter-bank theory [25], [26] supplies the PR conditions, the lifting scheme [23], [24] is the standard construction that satisfies them by design, and orthogonality has separately been used in CNNs to condition optimisation [29].

D. The closest prior result, engaged directly

WINNet’s Table II ablation compares decimated and undecimated Haar, DCT of several sizes, and learned Cayley operators on natural-image AWGN denoising, and reports that the learned operator performs similarly to a fixed DCT — whereupon they default to the fixed one. This superficially resembles our negative result, so we address it here rather than leave it for a reviewer. Two differences matter. First, their learned operator is constrained to be orthogonal, so PR holds throughout: their comparison is between two invertible transforms, whereas ours is between invertible and non-invertible ones — the failure we diagnose is not reachable in their design, and neither paper reports a round-trip measurement anywhere. Second, theirs is a default-picking ablation without significance testing; ours is a five-seed paired comparison with confidence intervals.

E. Our position

We propose neither a new wavelet nor a new architecture — the lifting construction in Section III-B is LINN’s and WINNet’s, and WINNet already learns the filter taps while preserving PR. What is new is the missing control. Prior invertible-wavelet denoisers assume structural invertibility and never test its absence, and so cannot report what happens without it. Making a learnable transform’s invertibility optional, holding capacity and training fixed, and measuring the consequence — and the six-order-of-magnitude round-trip gap that explains it — is the contribution. Section V-B returns to the residual tension with WINNet’s ablation.

III. Method

A. Overview

The network operates in the wavelet domain:

$$x \xrightarrow{\text{PR-LWT}} \{\text{band}_b\}_{b=1}^7 \xrightarrow{\text{heads}} \{\text{band}_b - \delta_b\} \xrightarrow{\text{PR-LWT}^{-1}} \hat{x} \quad (1)$$

where x is the noisy input and $\delta_b = \text{head}_b(\text{band}_b)$ is a per-subband residual prediction. For 2 levels this gives seven subbands $\{\text{LL}_2, \text{LH}_2, \text{HL}_2, \text{HH}_2, \text{LH}_1, \text{HL}_1, \text{HH}_1\}$.

B. PR-LWT: a perfectly-reconstructible learnable wavelet

We use the lifting scheme. In 1-D, with even/odd splitting,

$$\begin{aligned} \text{Analysis: } d &= x_o - P(x_e), & s &= x_e + U(d) \\ \text{Synthesis: } x_e &= s - U(d), & x_o &= d + P(x_e) \end{aligned} \quad (2)$$

Crucially, synthesis reuses the same P and U as analysis. Invertibility is therefore a property of the structure, not of the learned values: for any P, U , $\text{reconstruct}(\text{decompose}(x)) = x$ up to floating point.

The 2-D transform is separable (lifting along height, then width). We initialize $P = [0, 1, 0]$ (identity prediction) and $U = [0, 0.5, 0]$, plus a $\sqrt{2}$ output scaling, so that the initialization is exactly orthonormal Haar (verified to 2×10^{-7} on all four subbands).

Parameter bounding is necessary, not optional. We parameterize

$$P = P_{\text{init}} + 0.5 \tanh(\theta_P), \quad U = U_{\text{init}} + 0.5 \tanh(\theta_U) \quad (3)$$

Without this, the taps drift to $|P| \approx 5$, at which point the float32 round-trip error degrades from 3.6×10^{-7} to 5.2×10^0 . Perfect reconstruction holds unconditionally in exact arithmetic; bounding is what makes it usable in float32.

Parameter count: 2 levels $\times$ 2 directions $\times$ 2 filters $\times$ 3 taps = 24.

C. Subband denoising heads

Each subband is processed by a lightweight head ($\text{conv}_{3 \times 3} \rightarrow \text{ReLU} \rightarrow \text{conv}_{3 \times 3}$, 16 internal channels) that predicts a residual (signed, no final activation). Seven heads contribute 2,135 parameters; the complete network has 2,159.

D. Training

Loss: L_1 between prediction and ground truth. Adam, lr 10^{-3} , 30 epochs, 128×128 random patches, batch size 8.

IV. Experimental Setup

A. Data

AAPM-Mayo 2016 Low Dose CT Grand Challenge [30], [31], 3 mm B30 paired subset: 10 patients, 2,378 quarter-dose / full-dose 512×512 slice pairs. Slices are paired by ImagePositionPatient[2] (z position); pairing by filename is not possible because the slice-index field is identical on both sides.

TABLE I
Patient-level splits. S2 aligns with the convention used by published work so that numbers are directly comparable.

Split	Train	Val	Test
S1 (main)	7 pat. (1,600)	L291	L506+L067 (435)
S2 (lit.)	8 pat. (1,943)	L067	L506 only (211)

TABLE II
Identity baseline, all ten patients. The spread (25.50–29.25 dB, a 3.75 dB range) is larger than any effect we report, so all results are reported per patient.

Patient	n	PSNR	SSIM	Appears as
L143	234	25.4991	0.7643	train
L310	214	26.2577	0.7273	train
L067	224	26.5576	0.7987	S1 test / S2 val
L109	128	26.6052	0.8165	train
L096	330	26.9097	0.7853	train
L291	343	26.9805	0.8088	S1 val / S2 train
L333	244	27.2508	0.8305	train
L192	240	28.2896	0.8344	train
L286	210	29.1479	0.8103	train
L506	211	29.2489	0.8759	test (both)
S1 test, combined	435	27.8630	0.8361	
S2 test (L506)	211	29.2489	0.8759	

B. Evaluation protocol

AAPM never defined a PSNR protocol (its official metric was radiologist reading), and at least five mutually incompatible conventions coexist on this dataset. We adopt the RED-CNN / CTformer lineage convention, used verbatim by six public repositories:

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preprocess   $x = (HU + 1024)/4096$  (no clipping)
evaluate    denormalize to HU; clip both prediction and
            reference to  $[-160, 240]$ ;
             $PSNR = 10 \log_{10}(400^2 / MSE_{HU})$ ,
            data_range = 400; SSIM per SSInyu/RED-CNN
aggregate   per-slice metrics, then mean

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Our implementation reproduces an independent measurement of the identity baseline exactly (L506 29.2489, L067 26.5576, combined 27.8630).

C. Splits

Two patient-level splits are used to show robustness (Table I).

D. The floor: doing nothing

Identity (output = input) is the floor every model must beat. All ten patients, sorted by PSNR, are given in Table II.

V. Experiments

A. Main results

Main results are in Table III.

The baselines marked ours were trained by us under the identical protocol — same splits, same patch size, batch size, optimiser, schedule and evaluation — so the comparison no longer rests on matching a published setup. Two observations follow.

First, an independent check of our protocol: on the literature-aligned split S2 our RED-CNN reaches 32.9471 dB on L506, within 0.02 dB of the independently published 32.93 dB for the same patient [1]. Our implementation and our evaluation convention therefore reproduce the reference result rather than merely resembling it.

Second, under that matched comparison, PR-LWT comes within 0.97 dB (S2) and 1.00 dB (S1) of RED-CNN while using $857\times$ fewer parameters (2,159 vs. 1,848,865). We state plainly that PR-LWT does not beat RED-CNN on PSNR; the claim under test in this paper is about what makes a learnable transform helpful, not about attaining state of the art.

B. Ablation: is invertibility the key?

Three arms, differing only in the wavelet (Table IV). The arms answer two distinct questions: pr vs. unconstrained isolates the effect of invertibility; pr vs. fixed isolates the effect of learnability.

Results (mean $\pm$ std over seeds; paired t -test with the training seed as the independent unit) are in Tables V and VI.

Three findings, identical on both splits:

- 1) Imposing perfect reconstruction makes the learnable wavelet significantly better than fixed Haar (+0.27 dB, $p < 0.001$).
- 2) It also makes it significantly better than unconstrained learning (+0.30 dB, $p < 0.001$).
- 3) Unconstrained learning is indistinguishable from fixed Haar ($p = 0.11 / 0.34$; the CI straddles zero on both splits).

Finding 3 is the key one. It is not an artefact of low statistical power, because the same experiment resolves a +0.27 dB difference at $p < 0.001$. Unconstrained learning genuinely buys nothing.

Relation to WINNet’s splitting-operator ablation. WINNet [20] reports the opposite conclusion for a superficially similar comparison: a learned, orthogonality-constrained splitting operator matched a fixed DCT on natural-image AWGN denoising, so they defaulted to the fixed one. Our setting differs along several axes at once — task (LDCT vs. natural-image AWGN), transform family (a 2-level lifting parametrization initialized at Haar, so learning departs from the fixed baseline rather than searching over all orthogonal matrices of a matched size), and evaluation (five-seed paired tests with confidence intervals vs. a default-picking ablation). We cannot attribute the discrepancy to any one of these from the published numbers alone, and we flag it rather than claim a general conclusion. What this experiment does establish is narrower, and we believe robust: within this architecture and task, the PR constraint is what makes learnability pay.

TABLE III

Main results. Rows marked ours, same split were trained by us under the identical protocol. Our RED-CNN reaches 32.9471 dB on L506, within 0.02 dB of the independently published 32.93 dB for the same patient — an independent check that our implementation and evaluation convention reproduce the reference result rather than merely resembling it.

Split	Method	Params		PSNR	SSIM
S1	identity	—		27.8630	0.8361
S1	PR-LWT (ours)	2,159	30.8314 $\pm$ 0.0815	—	—
S1	RED-CNN [1] (ours, same split)	1,848,865	31.8310	0.8824	—
S2	identity (L506)	—		29.2489	0.8759
S2	PR-LWT (ours)	2,159	31.9731 $\pm$ 0.0787	—	—
S2	RED-CNN [1] (ours, same split)	1,848,865	32.9471	0.9090	—
S2	RED-CNN [1] (as published)	$\sim 10^6$	≈ 32.93	—	—
S2	CTformer [4] (as published)	$\sim 1.4 \times 10^6$	≈ 32.9	—	—

TABLE IV

The three arms. Only the wavelet differs between them.

Arm	Wavelet	Learnable	Inv. guar.	P.
pr	lifting scheme	✓	✓	—
unconstrained	separate filt.	✓	×	—
fixed	orthonormal Haar	×	✓	—

TABLE V

Ablation results, PSNR in dB.

Split	Arm	PSNR	n
S1	pr	30.8314 $\pm$ 0.0815	5
S1	unconstrained	30.5341 $\pm$ 0.0305	5
S1	fixed	30.5591 $\pm$ 0.0256	5
S2	pr	31.9731 $\pm$ 0.0787	3
S2	unconstrained	31.6788 $\pm$ 0.0307	3
S2	fixed	31.6936 $\pm$ 0.0394	3

C. Mechanism: what unconstrained learning actually does

We measure the analysis–synthesis round-trip error of the trained transforms, on the final checkpoint of every seed (Table VII, and Fig. 1a).

Unconstrained learning drives the transform to a state where 12–15% of the signal is lost in a forward–backward pass (S1 per-seed range 0.11–0.15), against 1.6×10^{-7} — the float32 limit — for the two invertible arms: a gap of six orders of magnitude. The network must then denoise and compensate for its own transform.

The comparison above is correlational. The unconstrained arm both loses invertibility and performs like fixed Haar; that alone does not show the former causes the latter. We therefore intervene directly. Holding architecture, capacity and training protocol fixed, we inject a controlled round-trip error into the PR arm by having its synthesis step use $(1 - \varepsilon) \cdot U$ while analysis still uses U — reproducing, in isolation, exactly the defect the unconstrained arm develops on its own (Table VIII, Fig. 1b):

Performance falls monotonically with the injected error, and every dose is significant. At 8.7% round-trip error

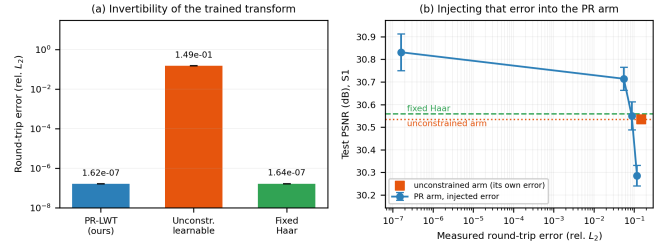


Fig. 1. (a) Round-trip error of the trained transforms ($n = 5$ seeds on S1; error bars are std): the two invertible arms sit at the float32 limit, the unconstrained arm six orders of magnitude above them. (b) Intervention: injecting that error into the PR arm degrades it monotonically, crossing both the fixed-Haar and the unconstrained reference levels. The square marks the unconstrained arm’s own operating point.

the PR arm lands at 30.5500 dB — on the fixed-Haar level (30.5591), the performance of an arm that never learned anything. The intervention therefore supports the mechanism causally: an error of this magnitude is not merely correlated with the failure, it is sufficient to produce it.

Honest boundary. The dose–response is somewhat steeper than the cross-arm comparison: at 11.8% error our injected arm reaches 30.2850 dB, below both the unconstrained arm (30.5341) and fixed Haar (30.5591). The unconstrained arm’s mismatch lies in both its analysis and synthesis filters, whereas our injection perturbs synthesis only — the two defects are of the same magnitude but not of the same structure. We therefore claim sufficiency at the same order of magnitude, not exact reproduction of that arm’s error.

PR-LWT keeps the round-trip error at the float32 limit while remaining fully adaptive. This is the entire mechanism.

VI. Discussion and Limitations

Why we do not claim “lossless”. Perfect reconstruction guarantees a round-trip only when the coefficients are unmodified. Denoising modifies them by definition, so PR does not imply a lossless denoiser. Our claim is narrower

TABLE VI

Paired comparisons. The unit of analysis is the training seed, not the test slice: the test sets contain 435 (S1) / 211 (S2) slices drawn from only 2 / 1 patients, and adjacent slices are highly correlated. Treating slices as independent would inflate significance by orders of magnitude.

Split	Comparison	Δ (dB)	t	p	95% CI
S1	pr – unconstrained	+0.2973	10.44	0.0005	[+0.218, +0.376]
S1	pr – fixed	+0.2723	10.68	0.0004	[+0.202, +0.343]
S1	unconstrained – fixed	−0.0251	−2.02	0.113	[−0.060, +0.009]
S2	pr – unconstrained	+0.2942	7.92	0.016	[+0.134, +0.454]
S2	pr – fixed	+0.2794	10.95	0.008	[+0.170, +0.389]
S2	unconstrained – fixed	−0.0148	−1.25	0.337	[−0.066, +0.036]

TABLE VII

Round-trip error (relative L_2) of the trained transforms, measured on the final checkpoint of every seed.

Arm	Wavelet params	S1	S2
pr	24	1.62e-07	1.65e-07
fixed	0	1.64e-07	1.64e-07
unconstrained	64	1.25e-01	1.45e-01

TABLE VIII

Intervention: forcing a controlled round-trip error into the PR arm. Every dose is significant (paired t -test, independent unit = training seed, $n = 5$). At 8.7% error the PR arm lands on the fixed-Haar level (30.5591 dB).

Injected ε	Measured RT	PSNR (S1)	Δ	Paired p
0 (control)	1.6e-07	30.8314 $\pm$ 0.0815	—	—
0.111	5.5e-02	30.7142 $\pm$ 0.0508	−0.117	0.0067
0.222	8.7e-02	30.5500 $\pm$ 0.0618	−0.281	0.0010
0.444	1.2e-01	30.2850 $\pm$ 0.0459	−0.546	<0.0001

and precise: the PR constraint removes the transform as a source of error the network must model, which is what allows adaptation to help.

Limitations.

- 1) Single dataset (AAPM-Mayo). Generalisation to other CT data is untested — including LoDoPaB-CT [32], which we note is a reconstruction benchmark (its observations are sinograms) and therefore not directly usable as a zero-shot denoising test without an FBP step.
- 2) Small test cohort (2 patients in S1, 1 in S2). This is why we report per-patient results and use seeds as the unit of analysis.
- 3) Same-split baselines cover RED-CNN only. RED-CNN was retrained by us under our own splits and protocol (Section ??); CTformer and the remaining methods are still quoted from the literature, and the training-set differences noted there apply to them. Retraining further baselines was out of scope here.
- 4) The injected defect is structurally simpler than the real one. We perturb the synthesis filter only, whereas the unconstrained arm’s two filter banks drift apart independently. The intervention establishes sufficiency at matched magnitude (Sec-

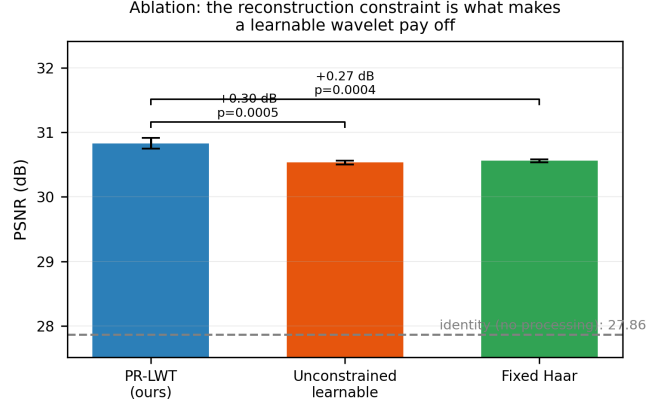


Fig. 2. Ablation: the reconstruction constraint is what makes a learnable wavelet pay off. Dashed line is the identity floor on the S1 set.

tion V-C), not exact reproduction.

- 5) No downstream task (e.g. segmentation) is evaluated.

VII. Conclusion

We asked whether making the wavelet learnable helps LDCT denoising. By default, it does not — an unconstrained learnable wavelet performs exactly like a fixed Haar basis, because learning destroys the transform’s invertibility. Constraining the wavelet to be perfectly reconstructible by construction — via a lifting scheme whose inverse shares the forward filters — makes adaptation pay off, for a network of only 2,159 parameters. The lesson generalises beyond this task: when a learned component is meant to be invertible, invertibility should be structural rather than hoped for.

Reproducibility

All numbers in this paper come from the 24 completed runs (3 arms $\times$ {5, 3} seeds $\times$ 2 splits) plus the two RED-CNN baselines and the identity baselines. Each run writes its full configuration (data sizes, parameter breakdown, LR milestones, code revision) to config.json and per-slice metrics to results.json. The round-trip measurements come from verify_roundtrip.py. Code and configurations will be released upon acceptance.

Visual comparison — patient L506, slice 100

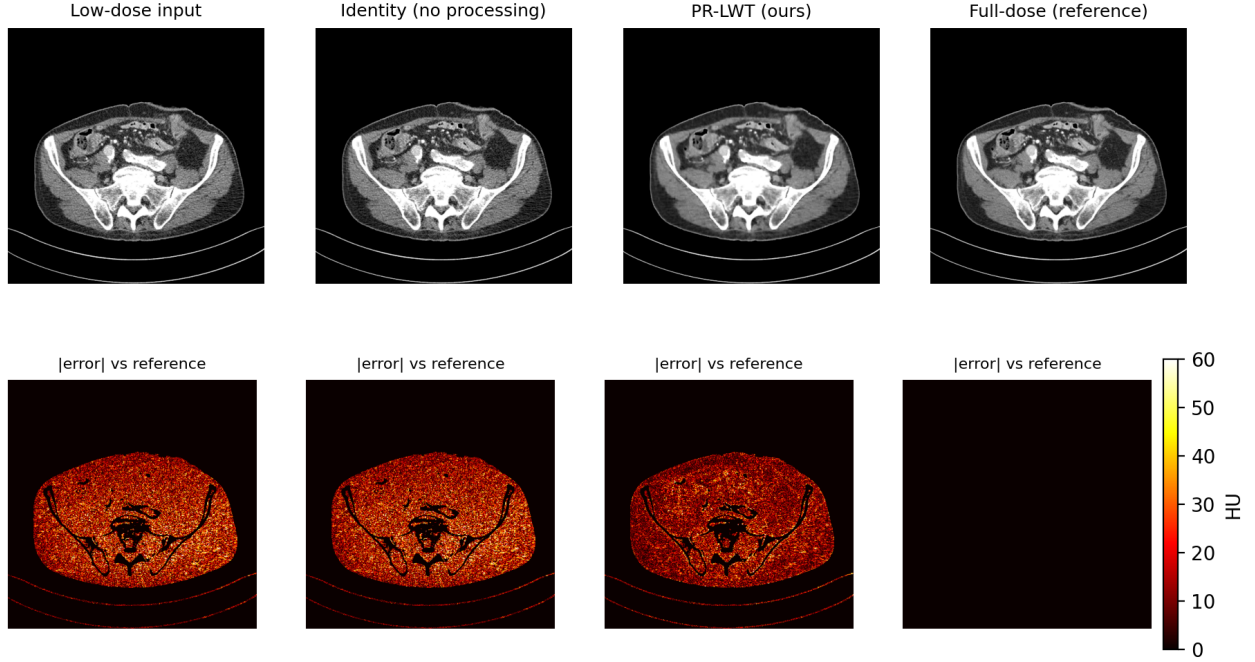


Fig. 3. Visual comparison on patient L506. Top: low-dose input, identity, our result, and the full-dose reference. Bottom: absolute error against the reference, in HU.

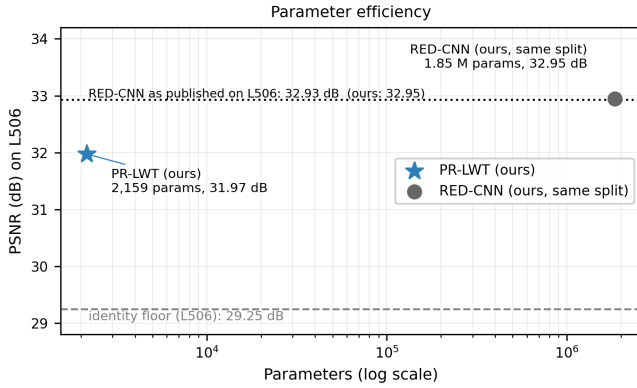


Fig. 4. Parameter efficiency on L506. The dotted line is the published RED-CNN score; our same-split reproduction lands within 0.02 dB of it.

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